SOUMIKA DAS

Software Engineer | Full-Stack & Backend Development | AI/ML

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Professional Summary

Computer Science undergraduate and Software Engineer with four internships across FinTech, HealthTech and research, building full-stack platforms, real-time systems and AI services. Published deep learning researcher and GDG speaker.

Education

B.Tech in Computer Science and Engineering (IoT, Cybersecurity and Blockchain Technology) Expected 2027
Institute of Engineering and Management, Kolkata GPA: 7.69

Core Skills

- **Languages:** Python, JavaScript, TypeScript, Java, C++, SQL
- **Frontend:** React, Next.js, HTML, CSS, Tailwind CSS, Vite
- **Backend:** Node.js, Express, FastAPI, Django, REST APIs, WebSocket, Microservices, JWT/OAuth
- **Databases:** PostgreSQL, MySQL, MS SQL Server, MongoDB, Vector DB
- **DevOps & Tools:** Git/GitHub, Docker, Linux, Nginx, CI/CD, PM2, Ansible
- **AI/ML:** GenAI, LLMs, RAG, Agentic AI, PyTorch, TensorFlow, OpenCV, NLP, CNN, LSTM, YOLO
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

BONAP - Botanical Scraper & Plant Portal | Live App

- Built a Node.js scraper and ETL pipeline extracting plant taxonomy, distribution maps and morphology from BONAP, USDA and GBIF, powering a Vite + React plant comparison portal with AI-generated summaries.

Brain Tumor Classification - Ensemble Deep Learning

- Built a CNN-LSTM ensemble classifying brain tumors from **3,200+ MRI scans**, reducing single-model variance.

DeepTrust - Degree Verification API | GitHub

- Built a degree verification API matching QR payloads to OpenCV/Tesseract OCR text with confidence scores.

Internship Experience

PepHub - Software Engineer Intern (Oct 2025 - Present) | Remote

- Delivered multiple client projects, including a Next.js/Node.js customer portal with **Stripe payments**, a FinTech REST API and a Vite botanical data tool, built on a modular, reusable architecture that reduced bugs.
- Automated production releases with PM2 and SSH, cutting deployment time by **82%** (45 min to 8 min).

IEDC, Dept. of CSE, UEM Kolkata - Research Engineer Intern (Dec 2024 - Dec 2025) | Hybrid

- Cut HPC cluster setup by **95%** (8 hrs to 22 min) with Ansible; built YOLO/OpenCV face detection at **96% accuracy**.

Cardiacrhythm - Software Engineer Intern (Jun 2025 - Sep 2025) | Remote

- Built WebSocket real-time ECG streaming for **200+ concurrent streams**, reducing latency by **90%** (1.8s to 180ms); shrank payloads by **75%** (8MB to 2MB) and added auto-reconnection to prevent data loss across **15K+ daily sessions**.

RevMaxx - Software Engineer Intern (Jan 2025 - Jun 2025) | Onsite

- Built a **HIPAA-compliant** transcription backend with SNOMED CT/ICD-10 coding and an NLP entity-extraction service across **50K+ clinical notes**; added retry logic and circuit breakers, cutting failures by **over 90%** (12/week to under 1).

Publications & Leadership

- **Brain Tumor Detection using CNN-LSTM**, AISC Conference (improved ensemble-learning performance) | LinkedIn
- **Cloud Lead & Speaker**, GDG on Campus UEM Kolkata (2025 - 26): led the Cloud domain, conducted Google Cloud Study Jams, spoke at “Tech and Adda” and co-organized the “StackUp!!” workshop. | LinkedIn